

Do Explanations Build Trust?

A Study on Explainable AI Recommendations

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Abstract

As artificial intelligence systems increasingly influence everyday decisions, users are often asked to trust recommendations without fully understanding why they were generated. This study examines whether explanations increase both subjective trust and actual decision-making behavior in an AI-based movie recommendation system. We designed and implemented an interactive prototype where participants evaluated movie recommendations under two conditions: without explanations and with explanations. The explanation condition included three explanation types: rating-based, similarity-based, and popularity-based explanations.

Twenty UIUC student participants completed the study. Each participant evaluated six movie recommendations twice, once without explanations and once with explanations, resulting in 240 total recommendation interactions. Trust was measured using a five-item Likert scale adapted from Hoffman et al. (2023), while behavior was measured through whether participants selected or rejected each recommendation. Decision time was also recorded automatically.

Results showed that explanations significantly increased trust, from a mean score of 2.32 without explanations to 4.27 with explanations, $t(19) = 19.39, p < .001$, Cohen's $d = 4.44$. Explanations also increased selection behavior, with selection rates rising from 35.8% to 90.8%, McNemar's $\chi^2(1) = 60.35, p < .001$. Similarity-based explanations produced the highest average trust descriptively, though explanation type differences were not statistically significant. Decision time did not significantly increase, suggesting that explanations improved confidence without adding meaningful cognitive burden. These findings show that explanations can strongly influence both perceived trust and actual user behavior, while also raising concerns about possible overtrust when explanations become too persuasive. The final version builds on the midpoint draft by adding completed participant results, statistical evaluation, and an interactive Streamlit dashboard for the final demo.

Keywords: explainable AI, trust, recommender systems, user behavior, human-AI interaction

1. Introduction

Artificial intelligence systems increasingly shape everyday choices, from movie recommendations to shopping, content, and services. As these systems become more common, a key question is whether users understand and trust the recommendations they receive. Without explanations, recommendations may feel random or difficult to evaluate; with explanations, users may better understand why a suggestion was made.

This project examines whether explanations increase trust in AI recommendations and whether that trust affects actual user decisions. We use movie recommendations because the domain is

familiar and low-risk, while still reflecting the structure of higher-stakes AI recommendation settings such as healthcare, finance, hiring, and education.

The final project builds on the midpoint prototype by adding stronger theoretical grounding, a validated trust measure adapted from Hoffman et al. (2023), completed participant data, statistical analysis, and an interactive dashboard demo. The main contribution is that the study measures both what users say and what users do by comparing trust scores with actual selection behavior in the same session.

2. Research Questions

This study is guided by three research questions:

RQ1: Do users report higher trust in AI recommendations when explanations are provided?

RQ2: Does the presence of an explanation increase the likelihood that users select the AI-recommended item?

RQ3: Do different explanation types, specifically rating-based, similarity-based, and popularity-based explanations, differ in their effect on trust and selection behavior?

3. Related Work and Theoretical Grounding

Explainable AI research argues that users need understandable reasons to evaluate AI outputs. Ribeiro et al. (2016) introduced LIME as a method for explaining individual model predictions in human-readable terms. Their work showed that explanations can help users identify when a model may be wrong, meaning explanations should not simply increase trust, but help users calibrate trust appropriately.

Lipton (2018) argues that interpretability is also an ethical issue. When AI systems influence decisions without explaining their reasoning, users lose the ability to evaluate or challenge the output. Lipton also distinguishes between transparent models and post-hoc explanations. This distinction is important for our study because our prototype uses structured metadata-based explanations rather than exposing a complex machine learning model's internal logic. This makes the system easier to control experimentally, but it also limits claims about full model transparency.

In recommender systems, explanations can serve several purposes. Tintarev and Masthoff (2007) identify goals such as trust, transparency, effectiveness, persuasiveness, efficiency, and satisfaction. Our study focuses mainly on trust and effectiveness. Trust is measured through participant ratings, while effectiveness is measured through whether participants actually select the recommendation.

Prior recommender-system research also shows that explanations can affect user behavior. Herlocker, Konstan, and Riedl (2000) found that explanation style can influence recommendation acceptance in movie recommender systems. Knijnenburg, Bostandjiev,

O'Donovan, and Kobsa (2012) similarly found that explanations can improve perceived system quality and intention to follow recommendations.

Recent work further supports the importance of measuring behavior, not just trust. Rahman, Siemon, and Ruotsalo (2025) show that persuasive explanations can influence users' choices in recommender systems. This directly supports our second research question, where we test whether explanations change actual selection behavior. Leichtmann, Humer, Hinterreiter, Streit, and Mara (2023) show that trust and human behavior do not always align in AI-assisted decision tasks. Their work supports our decision to measure both subjective trust and behavioral outcomes.

Finally, Hoffman, Mueller, Klein, and Litman (2023) provide validated measures for evaluating explainable AI systems, including trust, explanation satisfaction, mental models, and human-AI performance. We adapted our five-item trust scale from their framework to avoid relying on informal survey items. This strengthens the rigor of our measurement approach and directly addresses the feedback received after the midpoint submission.

4. System Design and Implementation

The final project includes a working Python-based recommendation prototype, results dataset, analysis notebook, and Streamlit dashboard. The system simulates a movie recommendation experience where users receive recommendations either with or without explanations.

Participants first enter preferences such as favorite genre, favorite movie, preferred year range, mood, and valued movie aspects. The system then presents movie recommendations under two conditions: no explanation and with explanation. In the explanation condition, users see one of three explanation types: rating-based, similarity-based, or popularity-based.

The implementation supports the full study flow: collecting preferences, presenting recommendations, recording trust scores, logging selection decisions, tracking decision time, exporting data, running analysis, and displaying results in the dashboard.

5. Method

5.1 Participants and Design

Twenty participants from the UIUC student community completed the study. The study used a within-subjects design, meaning each participant experienced both explanation conditions. Each participant evaluated six movie recommendations without explanations and six movie recommendations with explanations, producing twelve interactions per participant and 240 total recommendation interactions.

All participants completed the no-explanation condition first, followed by the explanation condition. This order was used to prevent explanation exposure from influencing the baseline no-explanation condition. However, this also creates a possible order effect, which is discussed as a limitation.

5.2 Measures

The study collected three main outcome measures: trust, selection behavior, and decision time.

Trust was measured using a five-item Likert scale adapted from Hoffman et al. (2023). Three items were common across both conditions: trust in the recommendation, perceived system reliability, and intention to reuse the system. Two condition-specific items were added to capture relevant aspects of each experience: the no-explanation condition included items on perceived match quality and confidence in the recommendation, while the explanation condition replaced these with items on understanding of the reasoning and helpfulness of the explanation. All items were rated on a 1–5 scale and averaged into a mean trust score.

Selection behavior was measured as a binary decision. Participants indicated whether they would select or reject each recommended movie. This captured actual behavioral response rather than only stated opinion.

Decision time was recorded automatically to evaluate whether explanations made the decision process slower or more demanding.

5.3 Analysis

For RQ1, a paired-samples t-test compared each participant's average trust score in the no-explanation condition against their average trust score in the explanation condition. Cohen's *d* was used to estimate effect size.

For RQ2, McNemar's test examined whether selection behavior changed when explanations were added. This test was appropriate because the selection decision was binary and paired.

For RQ3, explanation types were compared using trust and selection outcomes. A Friedman test was used for trust differences across explanation types, and a chi-square test was used for selection differences. Decision time was analyzed using a paired-samples t-test.

6. Results

6.1 RQ1: Effect of Explanations on Trust

Paired t-test revealed that explanations significantly increased trust ($M = 2.32$, $SD = 0.83$ without vs. $M = 4.27$, $SD = 0.56$ with explanations), representing a 1.95-point increase, $t(19) = 19.39$, $p < .001$, Cohen's $d = 4.44$ (very large effect). All 20 participants showed increased trust with explanations, demonstrating consistent directional effect (see Figure 1).

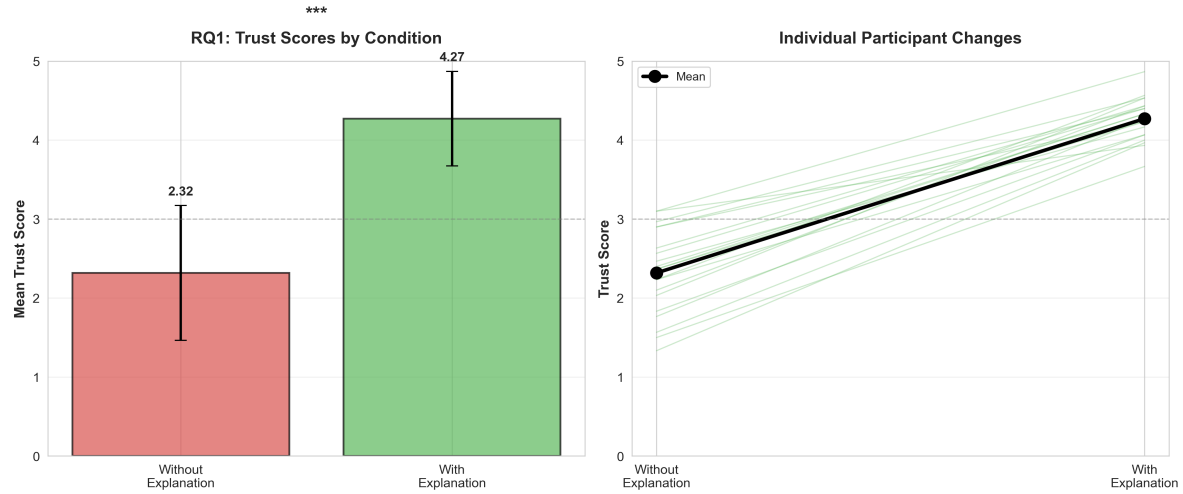


Figure 1: Trust scores by condition. Left panel shows mean trust with error bars and significance marking ($p < .001$). Right panel shows individual participant trajectories (all green = all increased)

6.2 RQ2: Effect of Explanations on Selection Behavior

Selection rates increased dramatically from 35.8% (43/120 movies) without explanations to 90.8% (109/120) with explanations, representing a 55-percentage point increase. McNemar's test showed 68 movies changed from not-selected to selected when explanations were added, while only 2 changed from selected to not-selected, $\chi^2(1) = 60.35$, $p < .001$, OR = 34.00. This odds ratio indicates movies were 15 times more likely to be selected when explanations were present (see Figure 2).

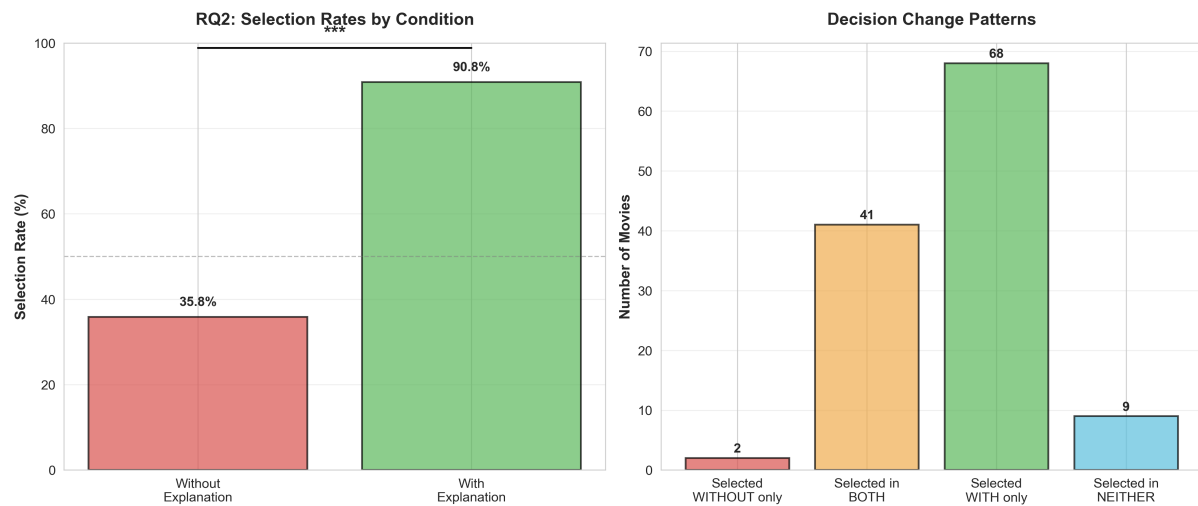


Figure 2: Selection behavior by condition. Left panel shows selection rates with significance marking. Right panel shows decision change patterns: 68 movies changed from not-selected to selected vs. only 2 in opposite direction.

6.3 RQ3: Explanation Type Effects

The three explanation types showed different descriptive patterns. Similarity-based explanations produced the highest average trust score, followed by popularity-based and rating-based explanations.

Explanation Type	Mean Trust	Selection Rate
Similarity-based	4.41	97.5%
Popularity-based	4.29	90.0%
Rating-based	4.13	85.0%

Although similarity-based explanations performed best descriptively, the statistical tests did not show significant differences between explanation types. The Friedman test for trust was not significant, $\chi^2(2) = 2.91, p = .234$. The chi-square test for selection behavior was also not significant, $\chi^2(2) = 3.80, p = .149$.

This means RQ3 is partially supported descriptively but not statistically. Similarity-based explanations appeared most effective, but the differences were not strong enough to conclude that explanation type had a statistically significant effect.

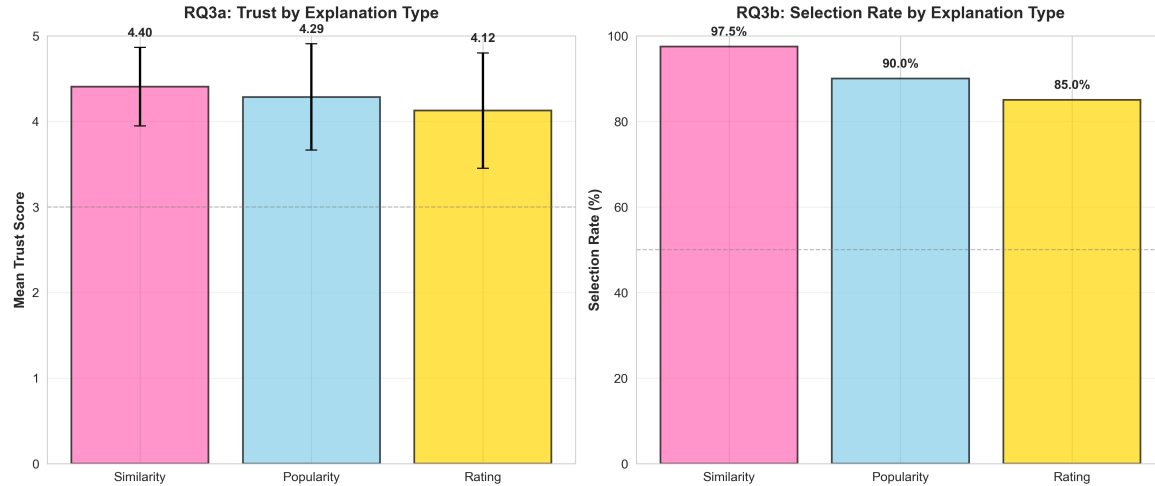


Figure 3: Trust and selection by explanation type. Similarity-based explanations showed the highest descriptive trust and selection rates, followed by popularity-based and rating-based explanations.

6.4 Decision Time

Decision time did not significantly differ between conditions. The average decision time without explanations was 6.72 seconds, while the average decision time with explanations was 7.05 seconds. This small increase was not statistically significant, $t(19) = 0.30, p = .769$.

This is important because explanations improved both trust and selection behavior without meaningfully slowing users down. In other words, explanations helped users feel more confident without adding a major time burden.

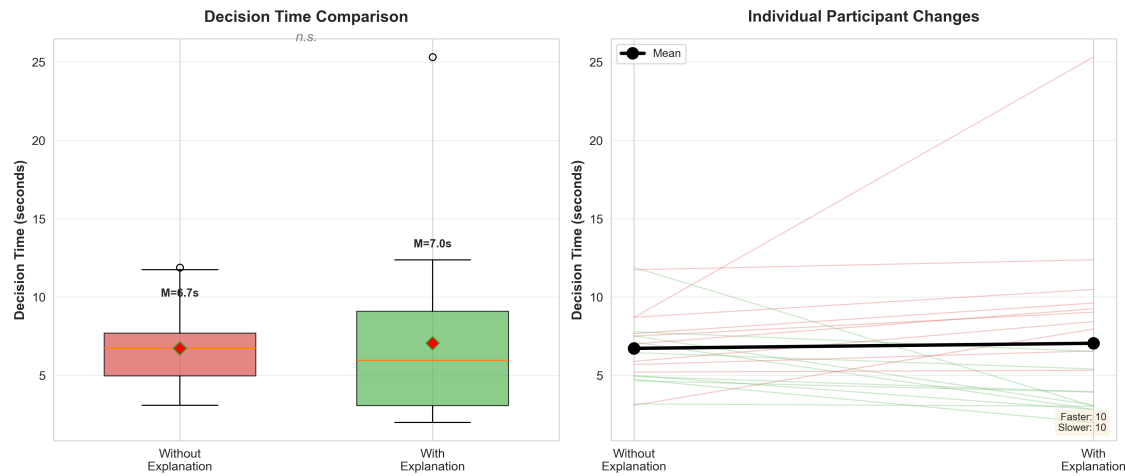


Figure 4 : Decision time by condition. Box plots show no significant difference between conditions (n.s.). Individual trajectories reveal balanced pattern: 10 participants faster, 10 slower with explanations.

7. Discussion and Reflection

The results show that explanations strongly influenced both trust and behavior in the AI recommendation system. Users trusted recommendations more when explanations were provided, and they were also much more likely to select those recommendations. The most important finding is that explanations did not only change perception; they changed action.

This finding connects directly to Rahman et al. (2025), who argue that explanations in recommender systems can be persuasive and influence user choices. Our study supports this with behavioral evidence. Participants were much more likely to accept movie recommendations after seeing explanations, suggesting that explanation design can directly shape decision outcomes.

The findings also connect to Leichtmann et al. (2023), who emphasize that trust and human behavior should be studied together. In our study, trust and behavior moved in the same direction. Participants reported higher trust and also selected more recommendations. This suggests that, in low-stakes domains such as movie recommendations, self-reported trust may be a useful predictor of behavior.

However, the strength of the effect also raises a concern. A selection rate of 90.8% with explanations suggests that explanations may strongly persuade users. This can be helpful when recommendations are accurate and explanations are faithful, but it can become risky if explanations sound convincing while hiding weak reasoning. This connects to NIST's (2021) idea of calibrated trust: the goal of explainable AI should not be to maximize trust blindly, but to help users trust systems appropriately.

Similarity-based explanations produced the highest trust and selection rates descriptively. This likely happened because similarity explanations connect directly to the user's own preferences. A user may find "this matches your favorite genre and mood" more personally meaningful than "this movie is highly rated" or "this movie is popular." Still, the explanation type differences were not statistically significant, possibly because the sample size was small or because all explanation types were persuasive enough to influence behavior.

There are also limitations. The study used a small convenience sample of 20 UIUC students, so the findings may not generalize to broader populations. The movie domain is low-stakes, so user behavior may differ in higher-stakes domains like healthcare, finance, or hiring. The explanations were based on structured metadata rather than a production-level black-box model, which improves consistency, but limits claims about full AI transparency. Finally, the no-explanation condition always came first, which may introduce order effects.

Future work should test explanations in higher-stakes domains and examine whether explanations help users detect poor recommendations or simply make them more likely to accept any recommendation. Future studies could also test adaptive explanation systems that choose explanation types based on user preferences, uncertainty, or decision context.

8. Conclusion

This study provides empirical evidence that explanations in AI recommendation systems meaningfully increase both perceived trust and actual decision behavior. By measuring both outcomes within the same session, we demonstrate alignment between subjective trust and behavioral choice, addressing a critical gap in XAI literature. The finding that explanations achieve these benefits without decision time costs suggests they reduce uncertainty rather than imposing cognitive burden. As AI systems increasingly mediate consequential decisions, understanding when and how explanations genuinely influence behavior becomes essential for responsible AI deployment. Our work demonstrates that well-designed explanations do more than make users feel better, they change what users choose.

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